MTH 107(7 and 10) Dr. James Ashe
Quiz 4.4 part 2 Friday 14^{th} November, 2014 **Name:**

Unless stated otherwise, all problems are worth equal value and answers should be stated as decimals rounded to three decimal places. You must show your work to receive credit.

1. Find c such that each of the following is true.

a. $p(z < c) = 36.69\%$

$c =$

b. $p(z < c) = 89.5\%$

c. $p(z > c) = 25.78\%$

d. $p(-c < z < c) = 81.64\%$

2. A population is normally distributed with a mean 60 and a standard deviation of 15. Find c such that each of the following is true

a. $p(x < c) = 36.69\%$

b. $p(x < c) = 62.93\%$

3. The time it takes employees to packag the components of a product is normally distributed with $\mu = 8.5$ minutes and $\sigma = 1.5$ minutes. Managment has decided to retrain the 35% of employees who took the greatest amount of time to repackage the components. Find the amount of time taken to package the components that will determine if an employee will be retrained.